

SUCCESSFUL CLINICAL AUDIO–METADATA ALIGNMENT DOES NOT IMPLY DISEASE TRANSFER: A PAIRING-CONTROLLED AUDIT

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ABSTRACT

Clinical audio models increasingly align recordings with participant metadata, but successful correspondence does not establish a transferable disease representation. Metadata mixes disease-related information with demographics, symptoms, history, and cohort structure. We introduce a Pairing-Controlled Transfer Audit comparing correct pairs, metadata shuffled within disease labels, and globally shuffled metadata. On UKCOVID, Correct pairing improved profile retrieval for AST-6L and OPERA-CT and retained more sex information than Within-label ($\Delta\text{AUROC} +0.191$ and $+0.112$), confirming that exact pairing changed the representation. Matched COVID prediction changed by only $+0.006$ and -0.002 . Across secondary analyses on CODA TB, Cambridge COVID-19 Sounds, and Coswara, retrieval and sex decodability increased consistently, whereas 0/11 dataset–backbone settings established a positive matched-disease gain, and Correct did not consistently outperform frozen raw audio. Correspondence learning, retained participant information, and disease transfer are therefore distinct empirical claims requiring distinct controls.

Index Terms— clinical audio, audio–metadata alignment, contrastive learning, confounding, domain shift

1. INTRODUCTION

Respiratory audio models increasingly combine cough, breathing, or lung sounds with patient information such as age, sex, symptoms, smoking status, and medical history. General-purpose representations including AST, OPERA, and HeAR provide strong audio encoders [1, 2, 3], while metadata has also been used as supervision for respiratory contrastive learning [4]. More recent multimodal systems explicitly align each recording with clinical text from the same participant. RespiraMFM, for example, freezes audio and text encoders and trains an audio-side projector before downstream multimodal prediction [5].

This alignment creates an evaluation ambiguity. A contrastive objective rewards any audio feature that helps recover the assigned metadata profile. Clinical metadata contains not only disease-related information but also demographics, symptoms, medical history, missingness patterns, and variables associated with cohort composition. Successful alignment can therefore reflect participant-profile correspondence, label-conditioned population association, or disease information that remains useful after population shift. Alignment loss, profile retrieval, or source-cohort AUROC alone cannot distinguish them.

The missing control is the *pairing relation itself*. Comparing correctly paired metadata only with globally shuffled metadata changes both exact correspondence and label-conditioned statistics. We introduce a *Pairing-Controlled Transfer Audit* with three otherwise identical arms: Correct preserves the observed pair,

Within-label assigns metadata from another participant with the same disease label, and Global removes systematic participant- and label-level pairing. Correct-minus-Within ($C - W$) asks what exact profile pairing adds beyond label-conditioned association, whereas Within-minus-Global ($W - G$) measures the latter. These contrasts are descriptive, not causal estimands.

The audit asks three sequential questions: *Was pairing learned?* *What information was retained?* *Does that information transfer?* Duplicate-aware profile retrieval is a manipulation check; probes identify decodable channels; and participant-disjoint, covariate-balanced disease evaluation tests transfer against Within-label and frozen raw audio. Across four datasets, Correct repeatedly improves profile correspondence and, most consistently, sex decodability. Its disease effect is small, directionally inconsistent, and does not consistently exceed raw audio. Successful correspondence and transferable disease representation are therefore separate empirical claims.

2. DATA AND METHODS

2.1. Datasets and evidence hierarchy

UKCOVID is the discovery dataset [6]. After objective audio filtering, its Standard train contains 20,714 participants. We evaluate Standard test ($n = 11,119$), a covariate-matched test ($n = 1,814$), and a participant-disjoint matched-long sensitivity set ($n = 4,196$). Recruitment source alone predicts COVID with AUROC 0.9966 in Standard train but 0.5000 after matching. This sharp shift is consistent with prior evidence that COVID audio performance can collapse after symptom and demographic adjustment [7]. Because UKCOVID tests informed earlier protocol development, we label it an exploratory discovery audit.

Three external analyses test whether the result shape recurs. CODA TB includes 1,081 eligible participants; we use a frozen 200-participant (100-pair) matched target from its released training data [8]. Cambridge uses 989 participants and a 200-participant target under a predefined strict-COVID reconstruction of its Task-2 cough subset [9]. Coswara uses 1,701 participants and a 200-participant target obtained by a model-blind secondary matching procedure [10]. CODA is a match-first secondary analysis, Cambridge is not an official benchmark replication, and Coswara is a post-hoc stress test. Matching balances prespecified measured covariates; it does not eliminate unmeasured confounding.

2.2. Representations and pairing interventions

Frozen AST-6L and OPERA-CT are the primary UKCOVID audio backbones; HeAR is added to the external backbone analyses. A frozen Phi-2 encoder [11] represents a deterministic schema containing available demographics, smoking, respiratory history, and

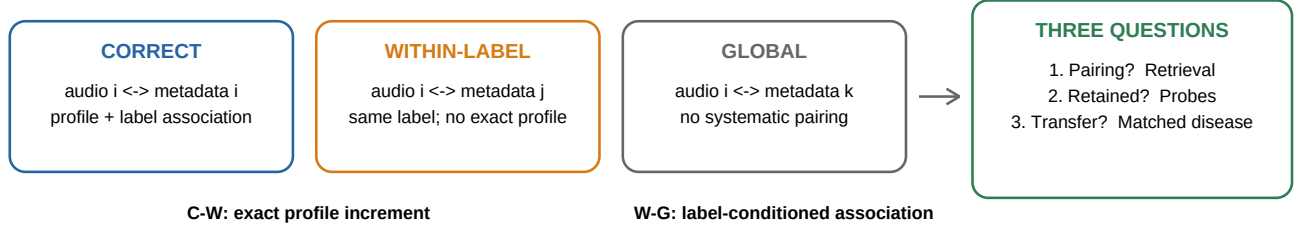


Fig. 1. Pairing-controlled audit. Correct and Within-label preserve the same label-conditioned metadata distribution in expectation, but only Correct preserves exact participant-profile pairing. Retrieval, probes, and matched disease evaluation answer three separate questions.

symptoms, with missingness explicit. Text excludes disease labels, participant identifiers, recruitment source, country/site, platform, device, timestamps, and recording artefacts. Thus, the risk is statistical association through patient context, not direct target or domain-label leakage.

For participant i , let a_i , m_i , and y_i denote audio, metadata, and disease label. The three training arms are

$$\begin{aligned} \text{Correct: } & a_i \leftrightarrow m_i, \\ \text{Within-label: } & a_i \leftrightarrow m_j, \quad y_i = y_j, \quad i \neq j, \\ \text{Global: } & a_i \leftrightarrow m_k, \quad i \neq k. \end{aligned} \quad (1)$$

Global removes systematic participant- and label-conditioned pairing, although accidental same-label or duplicate-profile matches can remain. The $C - W$ contrast is the increment from exact profile pairing after retaining label-conditioned metadata statistics in expectation; it is neither a pure identity effect nor a causal estimand. The $W - G$ contrast describes label-conditioned population association, and $C - R$ compares Correct with the frozen raw audio representation.

Only an audio-side projector is trained, using the one-directional audio-to-text InfoNCE objective from the audited Stage-1 implementation:

$$\mathcal{L} = -\frac{1}{N} \sum_i \log \frac{\exp(\bar{f}_\theta(a_i)^\top \bar{t}_i / \tau)}{\sum_j \exp(\bar{f}_\theta(a_i)^\top \bar{t}_j / \tau)}, \quad \tau = 0.07, \quad (2)$$

where bars denote ℓ_2 normalization and t_i is frozen text embedding. Each arm uses five seeds and 500 epochs; within a seed, arms share initialization, audio batches, and dropout streams, differing only in pairing. Only the final checkpoint is evaluated. Matched data select no epoch, projector, readout, or hyperparameter. This is a paper-faithful Stage-1 reimplement, not a reproduction of the full downstream Phi-2/LoRA system.

2.3. Audit endpoints and statistics

Correspondence: audio queries retrieve metadata profiles on source validation. Because multiple participants can share a schema, identical profiles count as the same semantic target; we report macro-profile MRR so frequent schemas do not dominate. **Information:** identical regularized probes decode sex, age, symptoms, acquisition/cohort variables, and disease. Decodability establishes information availability, not causal classifier use. **Transfer:** source data alone select and calibrate downstream readouts; paired AUROC, NLL, and Brier score are then evaluated once on matched targets. Intervals use participant-by-seed hierarchical bootstrap, retaining paired predictions.

Frozen raw-audio and metadata-only references check whether alignment improves over simpler representations. A fixed MLP checks linear-readout limitation. Finally, a one-dimensional calibrator fitted on participant-disjoint matched-long and transported to matched separates confidence transport from ranking.

3. RESULTS

3.1. Alignment occurred, but retained participant information

On UKCOVID validation, Correct improved macro-profile MRR over Within-label for both backbones. The gain was $+0.00318$ $[0.00161, 0.00492]$ for AST and $+0.00323$ $[0.00155, 0.00500]$ for OPERA. All five seed-level effects were positive. A null transfer result therefore cannot be attributed to a projector that learned nothing.

Correct also retained substantially more sex information on UK-COVID matched participants: ΔAUROC was $+0.1912$ $[0.1669, 0.2151]$ for AST and $+0.1125$ $[0.0943, 0.1311]$ for OPERA. Age increased for AST but not OPERA; acquisition and cohort effects were less consistent. Raw audio generally had higher absolute demographic decodability than aligned representations, so alignment did not create demographic information absent from audio; it selectively retained it relative to Within-label.

3.2. Correspondence did not robustly improve disease transfer

On UKCOVID matched, disease $C - W$ ΔAUROC was $+0.0062$ $[-0.0144, 0.0279]$ for AST and -0.0020 $[-0.0191, 0.0168]$ for OPERA. For AST, raw, Correct, and Within AUROC were 0.538, 0.529, and 0.523; for OPERA they were 0.577, 0.555, and 0.557. A small AST fusion effect did not replicate with OPERA, and Correct fusion did not outperform metadata plus raw audio. A fixed MLP also failed to reveal a hidden effect: audio-only $C - W$ was $+0.0050$ $[-0.0154, 0.0270]$ for AST and -0.0052 $[-0.0208, 0.0110]$ for OPERA.

Figure 2 summarizes the external result shape. Across all 11 dataset-backbone settings, retrieval point estimates were positive (10/11 intervals excluded zero) and sex $C - W$ was positive with 11/11 intervals excluding zero. In contrast, matched disease $C - W$ produced five positive and six negative point estimates, and 0/11 intervals established a positive gain. This count is descriptive, not a meta-analysis: targets contain different diseases, MRR libraries differ, and external analyses have different evidence status. Correct likewise showed no consistent advantage over frozen raw audio.

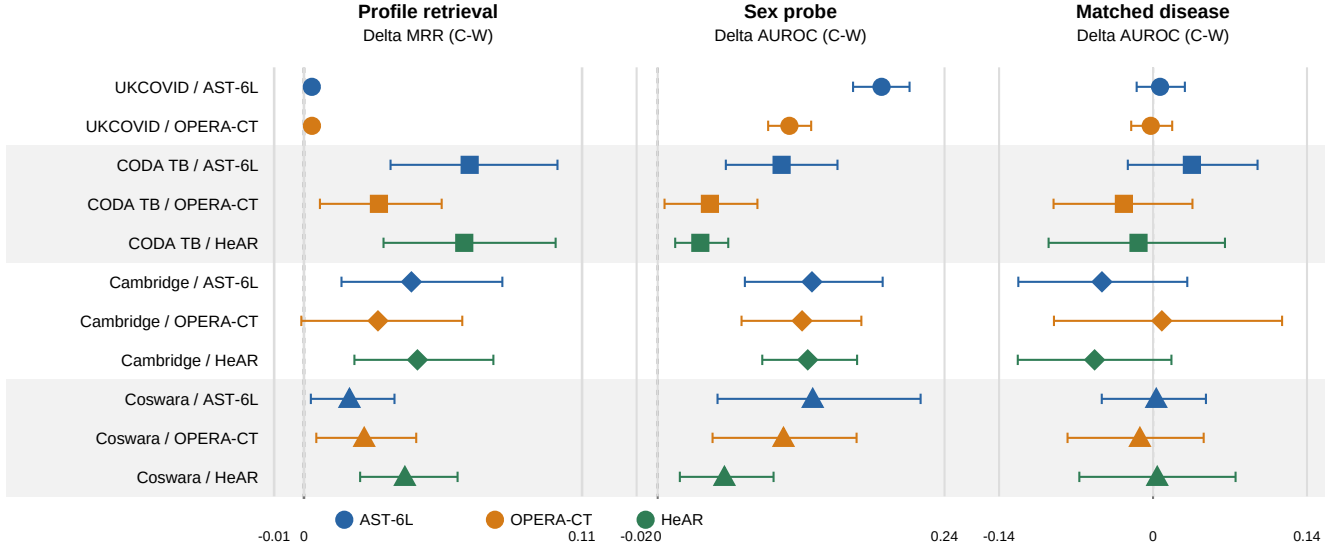


Fig. 2. Correct-minus-Within estimates with 95% paired bootstrap intervals. The 11 settings comprise UKCOVID (discovery: AST, OPERA) and three external sensitivities (AST, OPERA, HeAR each). Retrieval CIs excluded zero in 10/11 settings and sex-probe CIs in 11/11, whereas no matched disease CI established a positive gain. Panel scales differ; estimates are not pooled because profile libraries, diseases, and evidence status differ.

3.3. Label association, fusion, and calibration

Within-label and Global also separated in the source cohort. For UKCOVID AST, source $W - G$ probe Δ AUROC was +0.0646 for COVID, +0.0758 for cough, and +0.0647 for recruitment source; all three intervals excluded zero. On matched participants, the COVID increment fell to +0.0126 with an interval crossing zero, while cough remained positive (+0.0803). Thus, label-conditioned metadata association is itself learnable and would be conflated with exact pairing by a Correct-versus-Global comparison alone.

Direct fusion did not provide a robust alternative positive result. On UKCOVID matched, adding raw audio to metadata changed AUROC by only +0.0028 [-0.0016, 0.0073] for AST and +0.0041 [-0.0006, 0.0087] for OPERA. Metadata plus Correct exceeded metadata plus Within for AST (+0.0156 [0.0034, 0.0276]) but not OPERA (+0.0021 [-0.0068, 0.0107]); neither backbone’s Correct fusion beat metadata plus raw audio. This isolates a small, backbone-specific ranking signal rather than a reproducible alignment advantage.

With source calibration, UKCOVID Correct had worse matched NLL than Within, although AUROC differences were negligible. The change in negative NLL was -0.0249 [-0.0443, -0.0069] for AST and -0.0176 [-0.0335, -0.0010] for OPERA. Target calibration reduced both gaps to approximately zero without changing AUROC. The source NLL penalty therefore primarily reflects confidence unsupported after cohort shift, not demonstrated destruction of disease ranking; recalibration repairs probability scale but creates no transferable discrimination.

4. DISCUSSION

The audit separates two outcomes that conventional evaluation can conflate. Correct audio–metadata alignment repeatedly improved retrieval of participant profiles, proving that genuine correspondence was learned. Yet the most reproducible additional channel

was participant sex, while disease improvements were small, directionally inconsistent, and not robust to raw-audio controls or readout changes. In plain terms, the model became better at matching a recording to the kind of participant described by a metadata card, but not reliably better at diagnosing disease after measured patient composition was balanced.

Within-label shuffling is essential to this conclusion. Correct versus Global alone would mix exact profile correspondence with label-conditioned population association. Retrieval then verifies the manipulation, probes describe the retained channels, and matched evaluation asks the clinical transfer question. Future clinical audio–metadata studies should minimally report these controls together with a frozen raw reference and calibration under population shift.

This changes how positive alignment results should be interpreted. Low alignment loss or high profile retrieval establishes that paired supervision was absorbed; it does not say which part of a heterogeneous profile was used. A source-domain disease gain can then arise from label-conditioned patient composition, particularly when recruitment and label are coupled. Only a prespecified participant-disjoint target that balances measured patient context asks whether the gain survives the intended population change. Pairing is therefore a manipulation check, probes are representation diagnostics, and matched disease prediction is the transfer endpoint.

The recurring sex result also requires careful wording. A probe shows that sex is decodable, not that the disease classifier causally relies on it. Moreover, raw audio often contains more demographic information than any aligned representation. Correct therefore appears to preserve this channel relative to a Within-label projector rather than create a new demographic attribute. This distinction prevents a large $C - W$ probe effect from being misread as participant identification or newly introduced bias.

The evidence remains bounded. UKCOVID is a discovery audit; CODA, Cambridge, and Coswara are secondary or post-hoc sensitivities with 100-pair targets and wide intervals. Matching balances measured covariates only, probes do not establish causal use, and the

experiment audits one Stage-1-style projector rather than a complete multimodal reasoning system. Our synthetic study also failed its pre-registered general-mechanism gates, so these recurring patterns are not presented as a universal causal law. A proposed disease-invariant repair was not trained because the available single-dataset cohorts lacked the prespecified support for cross-environment same-label positives and within-environment opposite-label negatives. This is a data-feasibility limitation, not a negative result for the repair method.

5. CONCLUSION

Correct pairing reproducibly improved participant-profile correspondence, especially sex, but did not yield a robust cross-backbone improvement in covariate-balanced disease prediction or consistently outperform raw audio. Successful audio–metadata correspondence is therefore not sufficient evidence of a transferable disease representation; correspondence, retained information, and disease transfer must be evaluated separately.

Compliance with Ethical Standards

This work is a secondary analysis of de-identified datasets accessed and used under their respective data-use terms. No new participants were recruited and no clinical decisions were made from model outputs.

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